

3.5 Ordered Pairs, Cartesian Products

As stated previously, if a set A is given explicitly, it is immaterial in which order the elements of A are listed, e.g. the set $\{x, y\}$ is the same as the set $\{y, x\}$. In many instances, however, one would like, and, indeed, needs, to have some order in the appearance of the elements.

As an example, consider a coordinate plane with an x -axis and a y -axis; then we can identify any point in the plane by its coordinates (x, y) . If you wanted to find the point (a, b) , you would move on the x -axis a units to the right or to the left from the origin (depending on the sign of a), and then you would move b units up or down. If a and b are different, then (a, b) and (b, a) denote different points. So, in this example the order in which the elements appear is relevant. This leads to the following

Definition. Let A be a set;

1. A is called a *singleton* if $A = \{x\}$ for some x , i.e. if A has exactly one element.
2. A is called an *unordered pair*, if $A = \{x, y\}$ for some x, y , if A has exactly two elements.
3. A is called an *ordered pair* if $A = \{\{x\}, \{x, y\}\}$ for some x, y .

We shall usually abbreviate the right hand expression by

$$(x, y) = \{\{x\}, \{x, y\}\}.$$

The decisive property of ordered pairs is that two ordered pairs are equal if the respective components are the same. The following Theorem assures us, that the ordered pair as we have defined it has this property:

Theorem 1. *Let (a, b) and (c, d) be ordered pairs. Then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.*

Remark. The expression “if and only if” means that

1. If $(a, b) = (c, d)$, then $a = c$ and $b = d$.
2. If $a = c$ and $b = d$, then $(a, b) = (c, d)$. (This is called “the converse” of 1.)

So, we have to prove two directions, namely 1. and 2. Usually, “if and only if” is abbreviated as simply “iff”.

Proof. “ $\Rightarrow$ ”: Suppose $(a, b) = (c, d)$; then, by definition,

$$(a, b) = \{\{a\}, \{a, b\}\},$$

$$(c, d) = \{\{c\}, \{c, d\}\},$$

and since they are equal by our hypothesis, we have

$$\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}.$$

We consider two cases:

Case 1. $a = b$: Then,

$$(a, b) = \{\{a\}, \{a, b\}\} = \{\{a\}\} = \{\{c\}, \{c, d\}\},$$

Therefore, $\{a\} = \{c\}$.

Hence, $a = c$.

Also, $\{a\} = \{c, d\} = \{a, d\}$.

Hence, $d = a = b$.

Thus, for this case we have shown that $a = c$ and $b = d$.

Case 2. $a \neq b$: Then, $\{a\} \neq \{a, b\}$.

By our hypothesis we must have $\{a\} = \{c\}$, since

both sets have only one element; we get $a = c$.

Since $\{a, b\}$ has two elements, $\{a, b\} = \{c, d\}$.

Since $a = c$, $\{a, b\} = \{c, b\} = \{c, d\}$.

Therefore, $b = d$, and this is what we wanted to show.

“ $\Leftarrow$ ”: For the converse, let $a = c$ and $b = d$; then, $\{a\} = \{c\}$, and $\{a, b\} = \{c, d\}$, hence,

$$(a, b) = \{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\} = (c, d),$$

which is what we wanted to show. □

Definition . The *Cartesian (or cross-) product* $A \times B$ of two sets is defined as

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

So, the operation $\times$ pairs the elements of A with the elements of B in such a way that the elements of A appear as first components, and the elements of B appear as second components.

Note:

(i) $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$

i.e. for any object $\alpha \in A \times B$ if and only if $\alpha = (x, y)$

for some $x \in A$ and $y \in B$.

(ii) An element (x, y) of $A \times B$ is called the ordered pair, where x is designated as the first element and y as the second element. Specifically:

$$(x_1, y_1) = (x_2, y_2) \text{ if and only if } x_1 = x_2 \text{ and } y_1 = y_2.$$

In particular, $(x, y) = (x, y)$ unless $x \neq y$.

We write A^2 instead of $A \times A$.

(iii) $A \times \phi = \phi$.

Example:

consider $A = \{1, 2\}, B = \{a, b, c\}$.

Then: $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$.

And $B \times A = \{(a, 1), (b, 1), (c, 1), (a, 2), (b, 2), (c, 2)\}$.

Note that, $A \times B \neq B \times A$

Also, $A^2 = A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.

Theorem . Suppose A, B, C are sets . Then,

(i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

(ii) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

(iii) $B \subseteq C \Rightarrow A \times B \subseteq A \times C$

(iv) $(A \subseteq C \text{ and } B \subseteq D) \Rightarrow A \times B \subseteq C \times D$

(v) $(A \times B \subseteq A \times C \text{ and } A \neq \phi) \Rightarrow B \subseteq C$

(vi) $A \times B = B \times A \Leftrightarrow A = B$

Proof of (i)

Let α be an arbitrary element

$$\alpha \in A \times (B \cap C) \Leftrightarrow \exists x \in A \exists y \in B \cap C, \alpha = (x, y)$$

$$\Leftrightarrow \exists x \in A (\exists y, y \in B \cap C), \alpha = (x, y)$$

$$\Leftrightarrow \exists x \in A \wedge y \in B, \alpha = (x, y) \text{ and } \exists x \in A \wedge y \in C, \alpha = (x, y)$$

$$\Leftrightarrow \exists x \in A \wedge y \in C, \alpha = (x, y)$$

$$\Leftrightarrow \alpha \in A \times B \text{ and } \alpha \in A \times C$$

$$\Leftrightarrow \alpha \in (A \times B) \cap (A \times C)$$

$$\therefore A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Ex. Let A and B be nonempty sets. Prove that

01. $A \times B = B \times A$ if and only if $A = B$.

02. $A \times B \subseteq \wp(\wp(A \cup B))$.

End of D₁₇